Diffusion Policy · arXiv 2303.04137

Visionary

11+3 · seat 3

2026 control-loop bets

DP = Transformers-for-motor-control

Sep 21 paper club · VLA Architectures

Spine: chunk–commit–replan (T_p / T_a) — NOT classical planner

Chi et al. · RSS 2023 / IJRR extended

2026 control-loop bets

Control loop > bigger VLM alone for long-horizon fails

1 — Unify RTC + correlated inpaint

One recipe: async soft-inpaint with Σ-aware noise at train+infer

2 — Adaptive / event T_a

Gripper / contact triggers vs fixed 26/4

3 — Latency certificates

When is frozen prefix safe? DP §4.5 LQR is a seed

4 — Language basin selection

Which mode via language; DP-style commit inside chunk

ONE-LINER

DP is to continuous VLA motor control
what early Transformers were to NLP sequences —

the habit everyone still lives in after the backbone moved on.

Where DP still shapes 2026 robot learning

Action-head ontology

Discrete AR (FAST) = pretrain;
runtime dexterity wants
continuous generative chunks

Control loop > VLM size

BEHAVIOR fails on chunk jerks,
latency, mode flicker —
Fig 3/5 issues

Correlated noise

DP learned Σ implicitly;
winners put Σ in noise —
next: SE(3)/contact cov

Pose chunks lingua franca

Position/delta still dominate;
torque generative under-explored

Skip Discord/Drive

No product/community digressions
in this talk — control bets only

Infrastructure ask

Chunk generative + receding
exec = stack assumption,
not a paper trick

Club line to BEHAVIOR

DP chunk+replan → ACT → π_0 flow chunks → RTC soft-inpaint → BEHAVIOR (exec26/inpaint4)

DP
chunk + replan

ACT
CVAE chunks

π_0
flow chunks

RTC
soft inpaint

BEH
30/26/4 + Σ noise

What stayed vs what moved

STAYED
- generative continuous chunks
- execute-prefix / replan
- overlap consistency need

MOVED
- DDPM → flow matching
- sync → async soft-inpaint
- implicit corr → explicit Σ

Subsequent lineage · CLEAR citations only

Denoiser changed · chunk–commit–replan–overlap habit stayed

2023	2023	2024	2025	2025
Diffusion Policy	**ACT**	**π_0**	**$\pi_{0.5}$ / OpenPI**	**RTC + BEHAVIOR**
DDPM chunks T_p/T_a replan	CVAE chunks (parallel)	cites Chi'23 DDPM→flow	same chunk habit	soft-inpaint 30/26/4

π_0 · 2410.24164

- Cites Chi et al. 2023
- Flow matching on PaliGemma
- Continuous chunks ~50 Hz
- Same control ontology

RTC · 2506.07339

- Soft-inpaint async chunks
- Freeze delay-safe prefix
- Works on diffusion OR flow
- No retrain

BEHAVIOR 1st · 2512.06951

- Pi0.5 + corr. noise
- predict 30 / exec 26 / inpaint 4
- q ~ 0.26
- NOT training DP itself

Chunk–Commit–Replan · the motion/control contract

NOT classical planner / Diffuser full-trajectory planning · actions-only, O-conditioned

OBSERVE

$T_o \approx 2$

encode O_t once
(FiLM / cross-attn)

PREDICT

$T_p \approx 16$

denoise $A^k \rightarrow A^0$
joint action chunk

COMMIT

$T_a \approx 8$

execute prefix only
then stop

REPLAN

shift + loop

new O_t · warm-start
optional

One control cycle (@ ~10 Hz cmds → 125 Hz interp)

T_a commit (~0.8 s) | T_p prediction horizon (full chunk) | discard / warm-start

$T_a = 1$

mode flicker / idle trap

$T_a \approx 8$

sweet spot (paper)

$T_a \approx T_p$

open-loop / sluggish

Multimodal basins · Langevin commits, then holds

Push-T left/right · Kitchen order · BlockPush stage order

Diffusion Policy

commits to ONE basin per rollout

temporal consistency across T_p

BET / BC-RNN (foil)

mode flicker mid-horizon

independent per-step GMM/k-means

H-loop · closed-loop replan vs open-loop chunks

Same failure mode as long VLA chunks without replan

CLOSED-LOOP

$T_a < T_p$ · replan every commit

1. O_t → denoise chunk
2. exec T_a prefix
3. new O_{t+ta}
4. replan (warm-start)

OPEN-LOOP

$T_a \approx T_p$ · rare / no mid replan

1. O_t → denoise long chunk
2. exec almost all of T_p
3. obs may have shifted
4. recovery fails / lags

Empiricist test: set $T_a = T_p$ and disable mid-episode replan; inject obs delay 0–4 steps

Expect clear gap at delay ≥2 — ancestor of VLA long-chunk brittleness

Talk arc · one-slide narrative

Archaeologist through-line for Sep 21

1 — Multimodal BC hacks
GMM · BET · IBC
fragile / slow

→

2 — DP LOCKS generative chunks+replan
score + T_p/T_a
contract

→

3 — VLAs swap flow for DDPM
π_0 / $\pi_{0.5}$
same habit

→

4 — RTC / BEHAVIOR
soft-inpaint +
correlated noise

HONEST READING

The habit of the control loop persisted.
The denoiser changed (DDPM → flow → soft-inpaint + Σ).

Do NOT claim BEHAVIOR trains Diffusion Policy itself — they train flow on Pi0.5.

Q & A

Control-loop bets for 2026

Diffusion Policy · Sep 21

Diffusion Policy: Visuomotor Policy Learning via Action Diffusion

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Abstract

This paper introduces Diffusion Policy, a new way of generating robot behavior by representing a robot's visuomotor policy as a conditional denoising diffusion process. We benchmark Diffusion Policy across 15 different tasks from 4 different robot manipulation benchmarks and find that it consistently outperforms existing state-of-the-art robot learning methods with an average improvement of 46.9%. Diffusion Policy learns the gradient of the action-distribution score function and iteratively optimizes with respect to this gradient field during inference via a series of stochastic Langevin dynamics steps. We find that the diffusion formulation yields powerful advantages when used for robot policies, including gracefully handling multimodal action distributions, being suitable for high-dimensional action spaces, and exhibiting impressive training stability. To fully unlock the potential of diffusion models for visuomotor policy learning on physical robots, this paper presents a set of key technical contributions including the incorporation of receding horizon control, visual conditioning, and the time-series diffusion transformer. We hope this work will help motivate a new generation of policy learning techniques that are able to leverage the powerful generative modeling capabilities of diffusion models. Code, data, and training details is available diffusion-policy.cs.columbia.edu

Keywords

Imitation learning, visuomotor policy, manipulation

(a) Explicit Policy (b) Implicit Policy (c) Diffusion Policy

Figure 1. Policy Representations. a) Explicit policy with different types of action representations. b) Implicit policy learns an energy function conditioned on both action and observation and optimizes for actions that minimize the energy landscape c) Diffusion policy refines noise into actions via a learned gradient field. This formulation provides stable training, allows the learned policy to accurately model multimodal action distributions, and accommodates high-dimensional action sequences.

1 Introduction

Policy learning from demonstration, in its simplest form, can be formulated as the supervised regression task of learning to map observations to actions. In practice however, the unique nature of predicting robot actions — such as the existence of multimodal distributions, sequential correlation, and the requirement of high precision — makes this task distinct and challenging compared to other supervised learning problems.

Prior work attempts to address this challenge by exploring different *action representations* (Fig 1 a) – using mixtures of Gaussians Mandlekar et al. (2021), categorical representations of quantized actions Shafiullah et al. (2022), or by switching the *the policy representation* (Fig 1 b) – from explicit to implicit to better capture multi-modal distributions Florence et al. (2021); Wu et al. (2020).

In this work, we seek to address this challenge by introducing a new form of robot visuomotor policy that generates behavior via a "conditional denoising diffusion process Ho et al. (2020) on robot action space", **Diffusion Policy**. In this formulation, instead of directly outputting an action, the policy infers the action-score gradient, conditioned on visual observations, for K denoising iterations (Fig. 1 c). This formulation allows robot policies to inherit several key properties from diffusion models – significantly improving performance.

- **Expressing multimodal action distributions.** By learning the gradient of the action score function Song and Ermon (2019) and performing Stochastic Langevin Dynamics sampling on this gradient field, Diffusion policy can express arbitrary normalizable distributions Neal et al. (2011), which includes multimodal action distributions, a well-known challenge for policy learning.

Cover · Chi et al. · RSS/IJRR

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